

Sadegh Etemad



Senior AI/ML Engineer | AI Technical Lead



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PROFESSIONAL SUMMARY

Senior AI/ML Engineer and Technical Lead with 8+ years of experience building and scaling production machine learning systems serving millions of users. Expertise in recommendation systems, spatiotemporal forecasting, LLM-powered applications, multimodal AI, and production-scale machine learning systems. Proven track record of leading end-to-end AI initiatives from research to deployment while remaining hands-on in architecture, model development, and MLOps. Passionate about delivering reliable AI solutions that drive measurable business impact.

EDUCATION

2015 – 2023

Ph.D. in Computer Engineering (Artificial Intelligence)

• Amirkabir University of Technology, Tehran, Iran

2013 – 2015

M.Sc. in Computer Engineering (Artificial Intelligence)

• Iran University of Science and Technology, Tehran, Iran

2009 – 2013

B.Sc. in Computer Engineering (Software)

• Shahrood University of Technology, Shahrood, Iran

WORK EXPERIENCE

March 2026 – Now

Senior AI Engineer at **CPI**, United Kingdom

- Designed the logical and physical architecture of the next-generation cancer diagnostic platform, enabling multimodal healthcare data integration and AI-driven diagnostics.
- Developed the initial AWS-based prototype of a multimodal AI platform for cancer diagnosis, enabling the integration of clinical, imaging, and molecular data.
- Defined end-to-end data flows and architecture standards for secure, interoperable, and production-ready AI solutions.

July 2025 – Feb 2026

Senior Research Engineer at **Teesside University**, Middlesbrough, United Kingdom

- Designed and implemented multimodal ML pipelines combining imaging, spatial transcriptomics, and metabolic data.
- Built deep learning models for cancer detection on large-scale histopathology and multiplex imaging datasets.
- Developed scalable single-cell processing pipelines for segmentation, feature extraction, and annotation.
- Applied transformer-based models and explainable AI techniques to real-world biomedical decision-support problems.

- July 2024 – July 2025* **AI Tech Lead (Hands On)** at **EdTech-Wallex**, Tehran, Iran
- Led the design and implementation of LLM-powered educational systems for university entrance exam (Konkur) preparation
 - Designed and deployed LLM-powered RAG systems used by tens of thousands of active students.
 - Built multi-chain LLM architectures for QA, MCQ generation, and personalized study materials in production.
 - Reduced LLM inference and content generation costs by ~15-20% through prompt optimization and routing.
 - Worked hands-on with LangChain, vector databases, and OpenAI-compatible APIs at production scale.
 - Collaborated with frontend and backend teams to productionize AI features.
- Jan 2023 – July 2024* **AI Tech Lead (Hands On)** at **Snapp!**, Tehran, Iran
- Led and owned ML decision-making for traffic prediction and ETA systems serving millions of daily trips across multiple cities.
 - Improved ETA prediction accuracy by ~8–12%, directly reducing ride cancellations and improving user satisfaction.
 - Designed spatiotemporal feature pipelines that reduced inference latency by ~30% under peak traffic loads.
 - Acted as final technical authority for ML architecture and production incidents, maintaining 99%+ system availability.
 - Scaled and mentored a team of 5 ML engineers, reducing model iteration and deployment cycle time by ~20%.
- May 2020 – Jan 2023* **Senior Machine Learning Engineer** at **Snapp!**, Tehran, Iran
- Built and owned traffic forecasting pipelines to improve ETA and routing accuracy at scale.
 - Improved routing stability during peak hours, reducing ETA variance by ~15
 - Deployed and monitored production ML models serving millions of requests per day with real-time observability.
 - Collaborated with backend and infra teams to ensure low-latency, reliable ML services in production.
- Dec 2018 – May 2020* **Machine Learning Engineer** at **Kian Digital**, Tehran, Iran
- Designed and deployed algorithmic trading systems combining quantitative models and machine learning.
 - Built and productionized financial forecasting models using LSTM and gradient-boosted trees.
 - Engineered trend detection pipelines on large-scale time-series data.
- Jan 2018 – Dec 2018* **Machine Learning Engineer** at **Digikala Company**, Tehran, Iran
- Implemented customer segmentation and personalization pipelines using RFM features and clustering models.
 - Delivered ML-driven insights that directly supported large-scale marketing and customer engagement systems.

SKILLS

- **Programming & Software Engineering:** Python (Expert), R, Golang (familiar), object-oriented design, scalable software architectures, and production-grade software development.
- **AI, Machine Learning & Foundation Models:** Machine Learning, Deep Learning, LLMs, RAG, Agentic AI, Multimodal AI, Generative AI, Transformer Architectures, Representation Learning, Graph Neural Networks (GNNs), Recommendation Systems, Computer Vision, NLP, Time-Series and Spatiotemporal Modeling.
- **AI Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, CrewAI, Scikit-learn, XGBoost, LightGBM, Optuna, Statsmodels.
- **Data Engineering & Storage:** PySpark, Pandas, NumPy, PostgreSQL, MySQL, MongoDB, ClickHouse, Cassandra, Redis, Qdrant, Faiss, Apache Kafka, RabbitMQ.
- **MLOps & Production AI:** MLflow, Airflow, DVC, FastAPI, TorchServe, TensorFlow Serving, Seldon, Feast, Prometheus, Grafana, OpenTelemetry, Jaeger.
- **Cloud & Infrastructure:** AWS (EC2, S3, IAM, Lambda, API Gateway, DynamoDB, Neptune, ECS, ECR, CloudWatch, Bedrock, Bedrock Agentcore, Sagemaker), Docker, Kubernetes, Argo CD, Git.

LANGUAGES

- Persian (Native)
- English (Fluent)